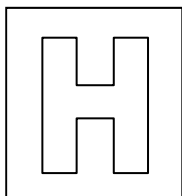


Candidate Name: _____

Class Adm No

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2018 Preliminary Exams Pre-university 3

BIOLOGY HIGHER 2

9744/04

Paper 4 Practical

20 September 2018

Candidates answer on the Question Paper

2 hour 30 minutes

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams and graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Answer all questions in the spaces provided on the Question Paper.

The use of scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	
2	
3	
Total	

This question paper consists of 19 printed pages, including 2 blank pages.

[Turn over

Answer **all** questions

1. The enzyme amylase catalyses the hydrolysis of starch to a reducing sugar.

You are required to identify which solution, **E1** or **E2** contains the highest enzyme concentration by estimating the concentration of reducing sugar produced by the action of **E1** and **E2** when breaking down starch.

The concentration of reducing sugar can be deduced by comparing with known concentration of reducing sugars. The reducing sugars are identified by using the Benedict's test.

You are provided with:

Labelled	contents	hazard	Volume/cm ³
E1	Amylase solution	Irritant Harmful	5
E2	Amylase solution	Irritant Harmful	5
S	Starch solution	None	50
G1	0.4% reducing sugar solution	None	50
W	Distilled water	none	100

Proceed as follows.

Read through step 1 to 7 and answer question a(i) to a(iii) before starting the experiment.

- Put 1 cm³ of **E1** and **E2** into separate beakers.
- Put 6 cm³ of **S** into each of the beakers with **E1** and **E2**. Mix well.
- Leave the beakers aside for 15 minutes.

During the 15 minutes, prepare the known concentrations of reducing sugar solutions according to **a(i)**.

(a)

- (i) Complete Fig. 1.1 to show how you will make three further concentrations of reducing sugar, **G2**, **G3** and **G4**. You should use serial dilution to reduce the concentration by half between each concentration.

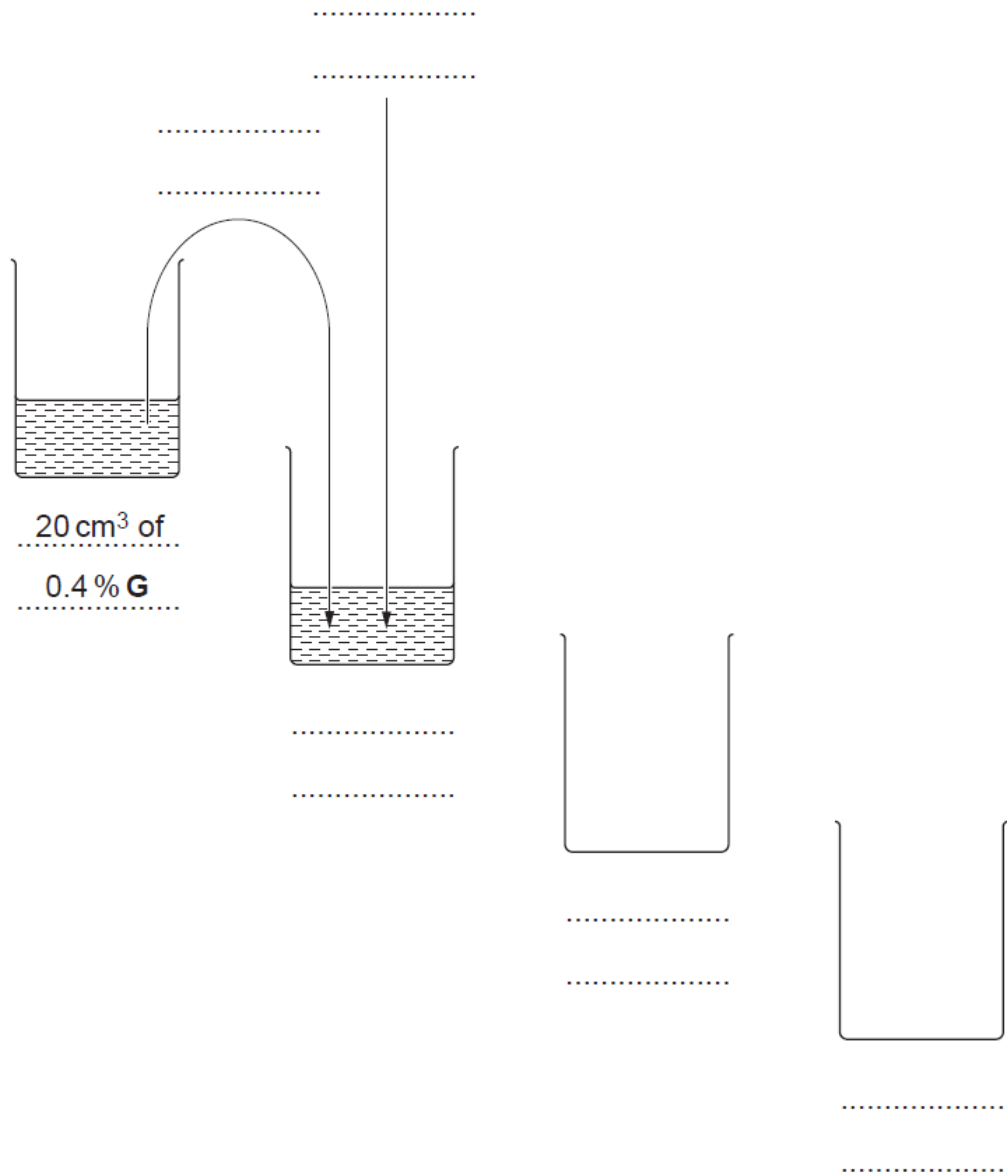


Fig. 1.1

[3]

- 4 Prepare all the concentrations of the reducing sugar solution which are stated in Fig. 1.1, in the beakers provided.
- 5 You are now required to test for the concentrations of reducing sugar by using the Benedict's test.

For each of your Benedict's test, you need to standardise the

- volume of Benedict's solution
- volume of the samples
- temperature of the water bath.

(ii) State the

Volume of each of the samples to be tested cm³

Volume of Benedict's solution cm³

Temperature of water bath °C

[1]

- 6 Set up the water bath to heat to the temperature as decided in a(ii).
- 7 Test each sample with Benedict's test separately using the volumes decided. Record the time taken for the appearance of any colour change. If there is no colour change after 120 seconds, record 'more than 120'.

(iii) Prepare the space below and record the results for Benedict's test.

[4]

- (iv) Using your results in **a(iii)**, state if **E1** or **E2** has higher concentration of reducing sugar.

..... **[1]**

- (v) Using your results in **a(iii)**, estimate the concentration of reducing sugars in **E1** and **E2**.

..... **[1]**

- (b)** Plan an experiment to follow the time course of the hydrolysis of starch by amylase, without the use of Benedict's solution.

You may select from the following sterilized apparatus and plan to use appropriate additional apparatus:

- Normal laboratory glasswares, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes and pipette fillers, glass rods, etc.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- Be illustrated by relevant diagrams(s), if necessary
- Identify the independent and dependent variables
- Describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- Use the correct technical and scientific terms.

[9]

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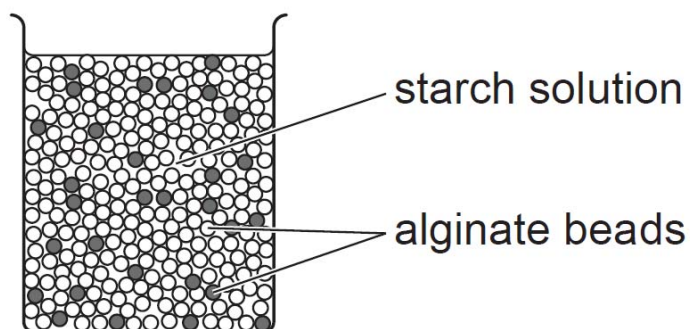
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- (c) In a separate experiment, the effect of iron sulfate on the rate of amylase activity was investigated. Amylase was immobilized in alginate beads. Two types of alginate beads containing amylase was prepared – with or without iron sulfate.

The two types of beads were mixed together in varying proportions. Fig. 1.2 shows an example of 30 beads with iron sulfate mixed with 70 beads without iron sulfate (30% with iron sulfate). The beads were placed into beakers containing starch solution.



Key:

- alginate beads with iron sulfate
- alginate beads without iron sulfate

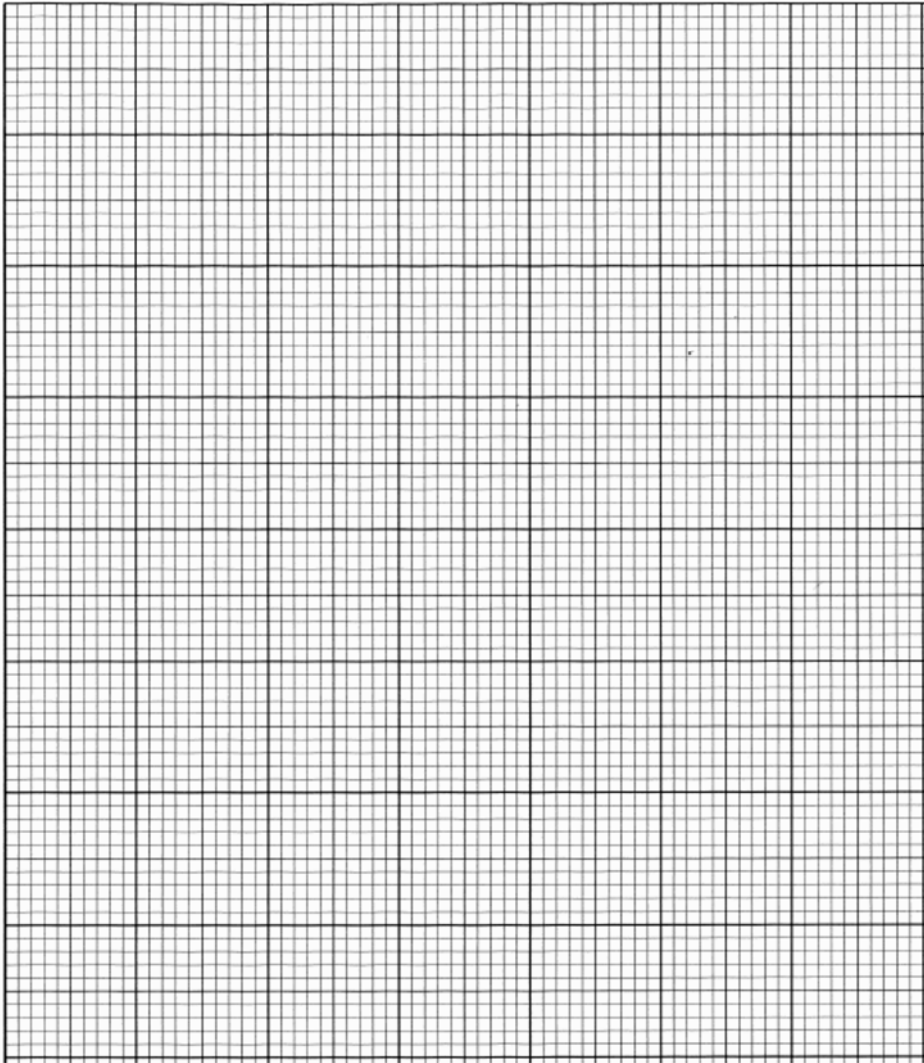
Fig. 1.2

The mass of reducing sugar produced by each mixture of beads in one minute was measured. The results are shown in Table 1.1.

Table 1.1

Percentage of beads with iron sulfate/%	Mass of reducing sugar produced/ μg
0	60
10	25
20	22
30	5
40	2
50	0

(i) Plot the data in Table 1.1 in the grid below.



[4]

(ii) State the anomalous data and explain your answer.

.....

 [2]

(iii) Describe two constant variables in this experiment.

.....

 [2]

- (iv) Explain the difference in the results between 0% and 10% of beads with iron sulfate.

.....

.....

.....

..... [2]

[Total: 29]

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2. You are required to investigate the light-dependent stage in photosynthesis in a leaf extract using the dye DCPIP. When DCPIP is reduced, it changes from blue colour to colourless.

You are provided with:

Labelled	contents	hazard	Volume/cm ³
L1	Fresh leaf	None	-
L2	Cold buffer solution	None	15
L3	Sand	None	20
L4	0.05% DCPIP	None	15

Proceed as follows.

- 1 Place the piece of fresh leaf, **L1**, on a white tile.
- 2 Using a scissor, carefully remove any large veins and discard them. Cut the rest of the leaf into small pieces and place them into a plastic container.
- 3 Using a pipette, add 3 cm³ of cold buffer solution, labelled **L2** into the plastic container.
- 4 Add a spatula of sand, labelled **L3** into the plastic container. Carefully grind the chopped leaf using a glass rod for about one minute to obtain a green leaf extract.
- 5 Clean the surface of the white tile and place a Petri dish on the edge of the tile as shown in Fig. 2.1. Ensure that the Petri dish is tilted.
- 6 Pour the leaf extract from the plastic container into the Petri dish and allow it to drain to the edge of the dish as shown in Fig. 2.1.

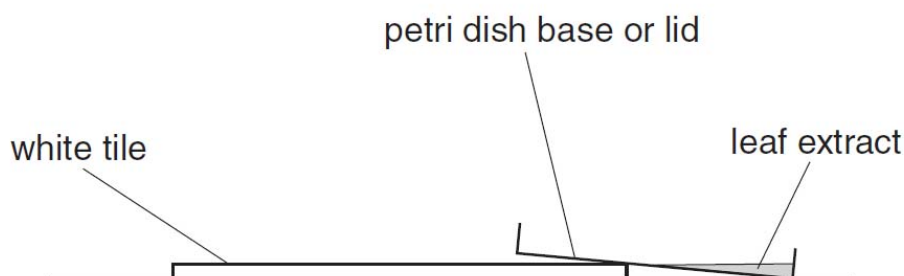


Fig. 2.1

- 7 Completely cover the Petri dish with aluminum foil to shield the content from light. Ensure that the foil is made easy to cover and remove.

You must cover the Petri dish with the foil at all times, except when adding reagent or removing samples.

- 8 Insert one end of a capillary tube into the leaf extract to collect some leaf extract. Remove the capillary tube and lay it on the white tile. **Label this tube as Tube 1.**
- 9 Using a pasteur pipette, add four drops of 0.05% DCPIP solution to the leaf extract in the Petri dish. Ensure the two liquids are evenly mixed to form a blue solution. If not, add more drops of 0.05% DCPIP to achieve a blue solution.
- 10 Take a second capillary tube and collect some leaf extract/DCPIP solution. Wrap this tube with aluminum foil to prevent any exposure to light. Place the capillary tube on the same white tile. **Label this tube as Tube 2.**
- 11 Take a third capillary tube and collect some leaf extract/DCPIP solution. Place the capillary tube on a separate white tile. Place the white tile and this capillary tube near a bench lamp. **Label this tube as Tube 3.**
- 12 Carefully check the colour of the liquid in all tubes every minute, for the next ten minutes. Note the time taken for the **DCPIP solution** to change from blue to colourless. If there is no change in colour after 10 minutes, stop the investigation.

- (a) Draw a table to show your results.

(b) Suggest the role of **Tube 1**.

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..... [1]

(c) Using your own knowledge, explain the expected results in relation to the light-dependent stage of photosynthesis.

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..... [4]

(d) Suggest the purpose of using the buffer solution, **L2** in step 3.

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..... [1]

(e) Describe two risks involved in this experiment and suggest how you could take precautions against them.

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..... [2]

- (f) Suggest three ways the investigation could be improved to make the results more reliable.

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..... [3]

[Total: 14]

3. Slide **S1** is a slide showing the transverse section of a plant stem. This plant is an important crop plant throughout the world.

- (a) Observe and select a group of four cells. Each cell in the group should touch two other cells.

Make a large detailed drawing of this group of four cells. Use one ruled label line to identify the component of the cell.

(b)

- (i)** Use the eyepiece graticule and the stage micrometer to measure the actual diameter of each of the four cells that you have drawn in **(a)**.

Identify each cell by writing the letters **A** to **D** on your drawing in **(a)**.

Show your working in the space below.

diameter of cell **A**

diameter of cell **B**

diameter of cell **C**

diameter of cell **D**

[5]

- (ii)** Calculate the mean diameter of cell **A** to **D** and the standard deviation. Use the formula for standard deviation.

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Show your working in the space below.

[3]**[Total: 12]****End of Paper**